

Prompt it Colorful with Rainbow Bot: Enhancing Video-Based Collaborative Learning by using a Multi-Party AI-Driven Chatbot

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Abstract

Multi-party AI-driven chatbots remain largely unexplored, despite their potential to enhance internal communication and coordination within student learning groups. These benefits are increasingly feasible with the advent of large language models (LLMs), which enable more efficient content generation and chatbot customization. To bridge this gap, we carried out an exploratory study to examine students' perceptions, practical applications, and concerns regarding the use of multi-party AI-driven chatbots in video-based collaborative learning environments. Based on the findings, we designed the Rainbow Bot, an optimized multi-party AI-driven chatbot. It features an interactive reflection process to strengthen group dynamics, customization options to adjust response length and content focus, and a unique color scheme for visual clarity. By combining colors from group members in refined prompts, Rainbow Bot fosters collaboration and promotes shared understanding within teams. Future work will assess its effectiveness in real educational settings, with a focus on its impact on group dynamics and collaborative learning outcomes.

CCS Concepts

• **Human-centered computing** → **Interactive systems and tools**; **Empirical studies in collaborative and social computing**; • **Computing methodologies** → **Artificial intelligence**; • **Applied computing** → **Collaborative learning**.

Keywords

Collaborative Learning, Multi-party AI-driven chatbots, AI-driven chatbots, Artificial Intelligence, Higher Education, Chatbot

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1 Introduction

Recent achievements in the field of artificial intelligence (AI) promise significant changes in many different areas, including education [21]. In particular, in the area of collaborative learning, recent research recommends that educational institutions foster environments that support group-based activities and collaborative projects through the use of AI tools [21]. By doing so, the potential of AI could be harnessed to improve team dynamics and cooperative collaboration among students. One promising approach involves the utilization of video-based collaborative learning environments, which enable participants to engage in real-time discussions and shared activities. These environments, augmented by AI tools, provide an interactive platform to possibly facilitate communication, engagement, and the overall effectiveness of group-based learning. For instance, chatbots, which have long been the subject of research in collaborative environments, can be employed to offer quick and targeted support and feedback [5, 6, 22]. In the past, these chatbots were predominantly examined in single bidirectional interaction patterns (one human and AI), while multi-party bidirectional interaction schemas (a group of humans and AI) were considered less frequently [10]. However, according to the shared mental model theory [4], achieving a shared understanding of the chatbot, as well as fostering internal communication and coordination among team members — which could be facilitated by a multi-party bidirectional interaction schema — might lead to improved team performance [17], team-skills [8, 23] and teamwork [2]. This, in turn, enhances the effectiveness of collaborative learning [26].

With the advent of large language models (LLMs), new possibilities for AI-driven chatbots in video-based collaborative learning contexts have emerged, offering more efficient content creation [20, 30] and support for diverse learning needs [9, 11]. In particular, the ability to customize the GPT system and thus the chatbot to the learning content is promising in order to provide students with targeted and precise answers from the AI. This customizability of AI can dynamically adapt to group dynamics [31], which could again promote a common shared mental model and its beneficial effects.

There is still limited research on multi-party bidirectional interaction schemes with AI-driven chatbots in video-based collaborative learning environments. To address this gap, we present an initial study exploring students' perceptions, identifying practical use cases, and addressing their concerns. This study aims to provide comprehensive insights into the role and effectiveness of AI technology in facilitating and enhancing video-based collaborative learning through collective use/interaction in multi-party teams.

Therefore, a field-study with 25 students has been conducted, in which students used an AI-driven chatbot integrated into an online video-based learning scenario to work collaboratively on lecture content. We afterwards conducted three focus groups to capture students' experiences. The results were clustered and resulted in mockups presenting an optimized AI-driven chatbot in collaborative video learning environments, called Rainbow Bot.

2 Related Work

Educational studies highlight the value of videos as effective tools for teaching and learning. Research by Merkt et al. shows that interactive video features, such as the ability to pause, enhance learning outcomes compared to traditional print media [19]. Further advantages of learning videos were also identified by Noetel et al., who found that videos used as supplementary tools in higher education significantly benefit students' learning experiences, while their replacement of traditional methods offers only modest improvements [24]. Beyond individual benefits, integrating collaborative elements into video-based group learning further amplifies these advantages. Studies by Brown and Lara demonstrate that collaborative learning not only boosts performance and productivity but also improves social skills, self-esteem, and group relationships [1]. Similarly, Laal and Ghodsi emphasize its social, psychological, and academic benefits, showing that group-based learning fosters success throughout education and beyond [13]. These studies make it clear that video-based collaborative learning environments have many positive effects on students.

These positive effects could possibly be further enhanced by AI tools to improve collaborative learning, particularly in video-based contexts. Mohebi et al. highlight how tools like ChatGPT enable seamless digital collaboration, supporting teamwork across physical distances [21] and fostering critical thinking through diverse perspectives and real-time feedback [27]. Li et al. also found that student teachers using ChatGPT in STEM training demonstrated improved critical thinking, efficiency, and reduced cognitive load, emphasizing the synergy between AI tools and human instructors [15]. One way to efficiently integrate AI tools into video-based collaborative learning is by employing AI-driven chatbots, which provide personalized learning experiences, enhance engagement, and reduce teacher workload, as noted by Orduño-Osuna et al. [25]. Yahaya et al. found these tools effective in virtual classes, with ease of use being crucial for adoption [29]. Kong et al. investigated an AI-powered discussion representation tool and found that the tool significantly improved higher-level cognitive presence and learning performance [12]. They encourage instructors to offer students visual representations or other supportive tools that can enhance the effectiveness of online collaborative learning. However, these chatbots are mainly utilized and researched in bidirectional interaction schemes, while their use in multi-party settings remains largely unexplored. In video-based collaborative learning environments, collaboration among students is central, with its positive effects on learning outcomes well-documented. Therefore, the use of multi-party AI-driven chatbots should also adopt a collaborative approach. For this reason this study represents an initial attempt to explore

multi-party AI-driven chatbots in video-based collaborative learning. *RQ1* investigates how participants in collaborative learning environments interact with multi-party AI-driven chatbots.

RQ1: How do multiple users interact with multi-party AI-driven chatbots in a video-based collaborative learning environment?

We aim to utilize multi-party AI-driven chatbots in a way that provides the greatest possible benefit to students, making it essential to understand how learners perceive these tools. We therefore assess participants' perceptions of multi-party AI-driven chatbots in video-based collaborative learning environments to understand what role participants assign to multi-party AI-driven chatbots in collaborative settings and whether there are differences based on the categorization from Deng and Yu (teaching assistant, learning partner or tutor) [7].

RQ2: What are learners' perceptions of multi-party AI-driven chatbots in video-based collaborative learning environments?

Identifying challenges and recognizing benefits can guide developers and educators: to address potential issues before implementation, and to maximize the positive impact on learning experiences. *RQ3* therefore focuses directly on the advantages and disadvantages of the usage of multi-party AI-driven chatbots in collaborative learning.

RQ3: What challenges and benefits arise from the usage of multi-party AI-driven chatbots in video-based collaborative learning environments?

3 Method

To address the research questions, we conducted a field experiment in an ongoing course with student participants from our university, who were given a voluntary option to participate. A total of 25 participants took part in the study ($N = 25$; 13 women and 12 men), who were divided into 7 groups, each consisting of 3–5 members. These groups had already been formed as part of the course, worked together throughout the entire semester, and were carried over for the study. The course was titled "MMI and GUI Programming", and the lecture topic, presented through the vGather2Learn tool in a video-based format [16, 28], was "Diversity, Inclusion, and Environmentally Conscious Design".

3.1 Experimental Environment

vGather2Learn is a video-based, collaborative learning tool integrated into the Moodle learning management system. It includes a video player for playing learning videos, a group-specific chat, features to create and join various individual rooms for groups within the tool, and functionalities to mark important and unclear points in the video using exclamation and question marks. For the present study, an AI-driven chatbot was integrated, which visually stands out from the regular chat through color differentiation. The ChatGPT API was integrated into the existing Node.js-based web application, utilizing the *gpt-3.5-turbo* model. The content for the AI was tailored to the learning material of the study session. Guidelines were established for the AI to answer students' questions based on the provided framework topics. These framework

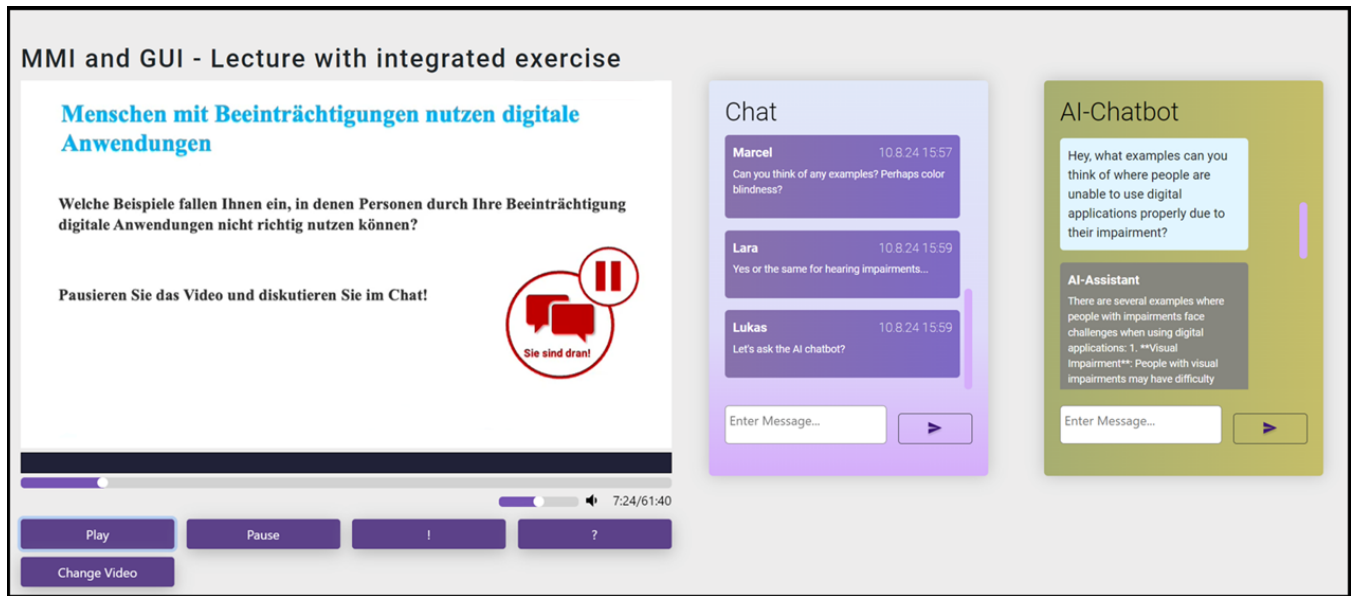


Figure 1: Interface of the vGather2Learn tool with the AI-chatbot.

topics, along with their corresponding timestamps from the video, were included in the content prompt. It was also instructed that if a participant asked for help in the context of the video, assistance would be provided directly with the relevant timestamp. We have decided to integrate the AI-driven chatbot as a separate instance in vGather2Learn alongside the existing chat. This ensures that conversations within the learning group remain separate from AI queries, preventing any interference with collaboration among group members.

3.2 Procedure

Participants registered for the study in advance. Pre-existing lecture groups were used for data collection, although the number of groups and group members varied, as participation was voluntary. All participants provided informed consent and attended a briefing explaining the study procedure. The study had two parts: first, each group watched a one-hour video with a collaborative learning tool featuring an AI chatbot. The video replaced a lecture and included three discussion questions for group debate. Participants communicated via chat or sought assistance from the AI chatbot. While the video content was relevant for the course exam, the discussion questions and experiment performance were not graded. Afterward, participants were divided into three focus groups, retaining their original course groups. The focus groups were divided into three blocks, each block representing a research question. Participants could use digital Post-Its, label them, and attach them to the corresponding question. This approach ensured that opinions were gathered from all participants with a low threshold for engagement. Group affiliations were color-coded to balance individual perspectives with group dynamics. Participants were also asked to justify their answers for deeper insights. The study concluded after the focus group discussions. All sessions were recorded, anonymized,

transcribed, and the Miro Board content was digitized and coded for analysis.

4 Results

To report the findings, the responses of the participants in the focus groups were categorized using a coding scheme by two researchers according to Mayring [18]. The coding scheme was developed inductively from the participants' answers, which were assigned to the categories of the generic terms "Person Perception", "Role of AI-Bot", "Advantages", "Disadvantages" and "Content of AI-Bot". These categories had subcategories, which are presented in the next subchapters. The subsequent results are reported according to their research questions. To assess intercoder reliability, Cohen's Kappa was calculated. The coders coded the answers of the participants on the Miro Boards. The Kappa value was $\kappa = 0.86$, which indicates almost perfect agreement according to the criteria of Landis and Koch [14]. In addition to the coding scheme, the logs of the video-based collaborative learning system were also examined and quantified. An overview of the categories can be found in Table 1. Beyond that, the participants' prompts were also examined in terms of content in order to draw conclusions about usage patterns. The results can be seen in the next subchapter.

4.1 RQ1: How do multiple users interact with multi-party AI-driven chatbots in a video-based collaborative learning environment?

Research question 1 was answered exclusively using the logs generated during the use of the collaborative video learning tool. The AI-driven chatbot was used 10 times to support **discussion questions** in the video, where the prompts reflected the questions in the video ("What are examples when people with disabilities cannot

Table 1: Statistics of the usage of the chat and AI-driven chatbot by participants

	Group A	Group B	Group C	Group D	Group E	Group F	Group G
Chats	95	13	52	7	0	5	13
Avg. Length	51.11	83.92	49.58	47.43	0.00	64.50	51.96
Total AI Requests	10	0	6	2	2	1	8
Discussion AI Requests	5	0	4	2	0	1	0
Other AI Requests	3	0	1	0	1	0	7
Organizational AI Requests	2	0	1	0	1	0	1
Number of persons used AI	5/5	0/3	3/3	2/4	2/3	1/3	3/4

use digital applications properly?"; Group A), 14 times for other **content topics** relevant to the lecture, such as terminology (e.g. "What is nudging?"; Group 2), or to simplify terms (e.g. "What is sustainability? Explain in simple terms"; Group 2), and 5 times for **organizational questions** about the course and the lecture that the AI chatbot could not answer ("Is the video available to watch later?"; Group G).

The number of individuals within the groups varied in their use of the AI-driven chatbot. In four groups, at least half of the participants interacted with the AI-driven chatbot, thereby fully realizing the multi-party concept. In contrast, the individuals in the other three groups participated less or not at all. Additional statistics can be found in Table 1.

4.2 RQ2: What are learners' perceptions of multi-party AI-driven chatbots in video-based collaborative learning environments?

4.2.1 Person Perception. A total of four categories were identified. The AI chatbot was described as **helpful** with 14 mentions. For example, Participant 19 said, "It summarized things, like from lectures and scripts, which would actually be helpful if you work on it in smaller parts." The AI chatbot was also regarded as **competent** with nine mentions. Participant 1 commented, "The AI thinks along and already knows what you want." With nine mentions, the AI chatbot was deemed **pragmatic**. Participant 4 noted, "Everyone sees the questions and answers so that no questions have to be asked twice". The AI chatbot was also perceived as **promoting** with 13 mentions. Participant 3 stated for example: "AI provides new food for thought - you can have questions from the lecture explained in greater depth".

4.2.2 Content of multi-party AI-driven Chatbots. When asked about the content multi-party AI-driven chatbots should provide in a university learning context, various possibilities were suggested. Five participants preferred that the AI be **limited to lecture content only**. Conversely, two participants expressed a desire for the AI to **offer additional, in-depth information** beyond the lecture content. **Definitions and examples** were mentioned three times each, while four participants wanted the AI to provide **summaries** upon request. Three participants wished for the AI to include interactive elements such as quizzes.

4.2.3 Usage of multi-party AI-driven Chatbots. Regarding how multi-party AI-driven chatbots should be used in a university learning context, nine participants suggested they be used as **supplementary aids**. Participant 15 says: "So the last option is simply to try it yourself first, but then, because you somehow don't get a result, simply ask the AI as a last resort.". Two participants believed they should **exclusively support discussions**, while another two participants thought the AI-driven chatbot should only be used as an **interactive element** e.g. with a quiz (Participant 1), the evaluation of solutions to tasks given to the chatbot by the participants (Participant 12) and the provision of frequently asked questions (Participant 13).

4.2.4 The Role of multi-party AI-driven Chatbot. Additionally, participants were asked in an open-ended question about the role the multi-party AI-driven chatbot assumed during the learning session. Here, 13 participants viewed the AI chatbot as a **helper and supporter**, three as a **source of ideas**, and two as a **new group member**. To explore the defined roles of chatbots [7], participants were asked to choose among the roles of learning assistant, tutor, and learning partner. Twenty-two participants chose learning assistant, while three chose tutor. Participant 1, who sees the AI assistant as a learning assistant, said: "So tutor doesn't work simply because you have to feed it with something and then it responds to it. It doesn't provide any content itself and says, do this and that. In other words, they always have to respond to the prompt. And learning partners, well, they don't really learn anything themselves.". In contrast, Participant 2, who opted for tutor, said: "Well, I just think a tutor can answer a lot of questions, but usually not everything. And the way we used it in our group, for me it was more of a tutor than a learning assistant.".

4.3 RQ3: What challenges and benefits arise from the usage of multi-party AI-driven chatbots in video-based collaborative learning environments?

4.3.1 Benefits of multi-party AI-driven Chatbots. Five categories of advantages were identified. **Efficiency** was the most frequently mentioned, with seven participants highlighting it as a benefit. For example, Participant 3 stated, "I think that answering questions benefits the whole group." One participant mentioned that using the chatbot **enhances understanding**. Participant 6 noted, "I find it easier for understanding, instead of just making a statement and

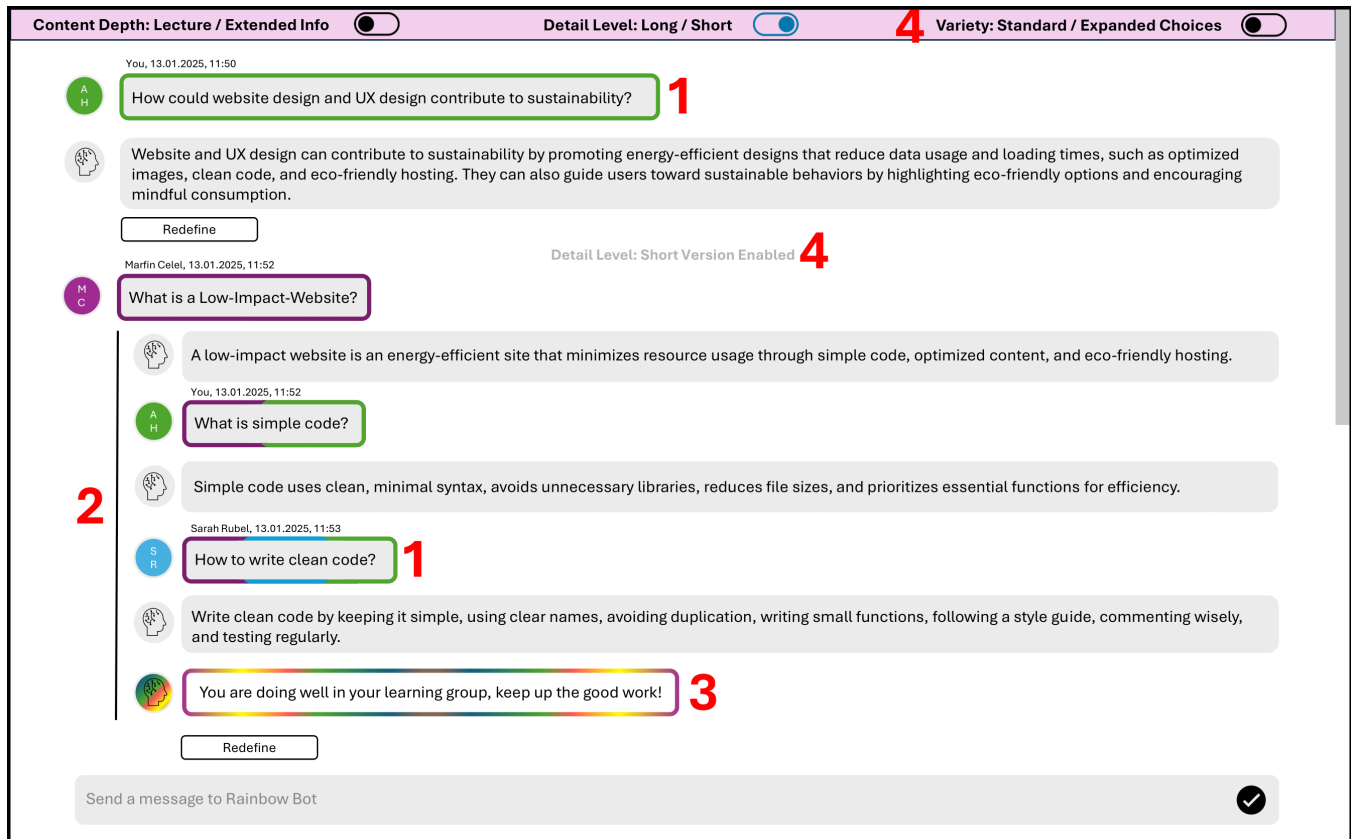


Figure 2: Mockup of the Rainbow Bot

saying, this is a fact. It allows for deeper exploration." Another advantage is the **additional support** provided by the chatbot (six mentions). Participant 14 explained, "I used it first to think about the problem myself, and then, when I was stuck, to see what the AI suggested." **Diversity of perspectives** was mentioned two times. Participant 6 said, "I found that it provided a different way of thinking and suggested alternative solutions." The last category identified was **facilitation of communication**, with two mentions. Participant 14 remarked, "In an online context, I would only consider writing to the professor if I had a very serious question. But with the chatbot, you can ask any kind of question."

4.3.2 Challenges and Disadvantages of multi-party AI-driven Chatbots. The challenges and disadvantages of multi-party AI-driven chatbots can be categorized into four main areas. The most significant drawback, with seven mentions, is **reduced individual initiative**. Participant 19 stated, "If you can just post the question and get an immediate answer that fits perfectly, you might not think about it yourself anymore." Three participants mentioned that multi-party AI-driven chatbots can lead to **information overload and overwhelm**, as expressed by Participant 9: "Yes, because you get so many perspectives at once, and it can be a bit overwhelming to keep track." Additionally, four participants believe that the use of AI in collaborative learning scenarios can result in **reduced group interaction**. Participant 2 commented, "In our discussion group, we

first asked AI what it thought about the question. This led to less interaction among us. And, for example, if there were unclear terms, we might have asked the group first if AI wasn't available." Finally, **unreliable** information was mentioned by two participants. Participant 13 remarked, "We asked two different questions twice and received the same answer both times. I don't know if AI thinks on its own yet, but I think the answers are limited or predetermined."

5 Discussion of the Evaluation

The chat log analysis highlights variability in group interactions with the multi-party AI-driven chatbot. While some groups actively engaged as a whole, there were also groups where only one or no participant contributed. The nature of the inquiries also varied across groups, ranging from questions about lecture content to organizational matters. These differences likely reflect variations in group dynamics, communication styles, and task structures. Participants' perceptions of the multi-party AI-driven chatbot also differed. Many viewed the chatbot as helpful, competent, and pragmatic, primarily identifying it as a "helper and supporter" rather than a substitute for a teacher. These perceptions align with prior research [3], which suggests AI tools are seen as complementary aids. The design of the AI chatbot was also criticized in terms of clarity, organization of elements and general modernity. Roles such

as "helper" and "idea provider" highlight a nuanced understanding of chatbots' functions, warranting further exploration of their impact on engagement and outcomes. Key benefits of multi-party AI-driven chatbots include improved efficiency, understanding, and communication. However, challenges such as reduced initiative, information overload, and diminished group interaction remain. Reduced initiative is particularly concerning, as it may limit the critical thinking potential that systems like ChatGPT are capable of fostering [27]. Addressing these challenges will require features that promote balanced use and active participation.

To enhance the effectiveness of collaborative learning, as outlined by the shared mental model theory [4], it is essential to find ways to improve collaboration within learning groups. We conclude that it is crucial to implement features that encourage active participation within groups and in the use of multi-party AI-driven chatbots. Additionally, ensuring that these chatbots can respond to and meet diverse needs is important. Such adaptations could support individual group dynamics and communication styles, ultimately leading to improved collaboration. An initial attempt to implement this functions resulted in the development of the rainbow bot.

6 Implications for the Rainbow Bot

We developed the **Rainbow Bot**, an optimized multi-party AI-driven chatbot, integrating insights from our results and additional innovative ideas (see Figure 2). Since the initial chatbot interface was described as lacking organization and clarity, the interface of the Rainbow Bot was adapted to resemble **common instant messaging tools**, enhancing user familiarity and engagement. In our multi-party interaction model, each participant is assigned a **unique color** that appears as a border around their messages (see Point 1 in Figure 2). This design visually distinguishes individual contributions and AI responses, promoting clarity and usability. Since we need to find ways to foster collaboration and improve shared mental models, as the individual use of the multi-party AI-driven chatbot varies, we introduced a **redefining function** inspired by thread features in instant messaging platforms (see Point 2 in Figure 2). This function allows students to collaboratively refine prompts for more precise, context-specific responses. The refined prompts visually combine the contributors' colors, emphasizing the iterative and collaborative nature of this process and showing visually that multiple users contributed to the improved query. We also integrated an **interactive reflection component** focused on group dynamics rather than content. For example, the bot might subtly encourage quieter members to participate or praise effective collaboration within the group (see Point 3 in Figure 2). These prompts are designed to be non-intrusive, fostering a supportive environment without feeling forced. As we have found that participants have different perceptions, expectations and purposes for using the multi-party AI-driven chatbot and to accommodate diverse learning needs, the chatbot offers **customizable settings**, enabling students to choose the **type of content**, **response length**, and whether to **include multiple perspectives** (see Point 4 in Figure 2). These settings are accessible via toggle buttons at the top of the interface, applying to all chatbot interactions. Additionally, users can override specific settings through individual prompts,

such as adjusting response length dynamically. This tailored approach leverages AI capabilities to support collaborative learning, enhance engagement, and effectively meet the varied needs of students.

7 Conclusion and Future Work

This paper examined students' perceptions, advantages, and concerns regarding the usage of multi-party AI-driven chatbots in collaborative learning environments. The results suggest that the use of AI chatbots in collaborative learning is perceived as a supportive tool, considered efficient and inspiring, but also criticized for reducing personal initiative. The usage logs of the AI-chatbots indicate that the utilization of such tools may be highly individual and group-dependent. Based on these results and own ideas, a redefined version, the rainbow bot, was created. The Rainbow Bot features a distinctive color scheme designed to enhance visual clarity and reinforce the collaborative aspect through its prompt-redefining function. Additionally, the Rainbow Bot offers various customization options for the chatbot and includes an interactive reflection component aimed at strengthening group dynamics. This study is not free of limitations. There were technical limitations in the provision of the AI by OpenAI. Since the GPT system was personalized to the lecture content, the maximum number of tokens that ChatGPT allows per session was quickly reached and resulted in the AI no longer processing any requests. The solution here was to greatly shorten the prompt that personalizes the GPT system and to limit the history track function. This meant that the GPT system only remembered the last prompts, which was sufficient in the context of the lecture. Moreover, in this experiment the groups used the AI-driven chatbot only for one lecture and not for a whole course. This should be addressed in future work. Regarding the Rainbow Bot, we plan to evaluate it in a university course, with a specific focus on how well the interactive functions and visual design align with our expectations of fostering collaboration within the groups.

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